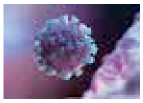
Detecting silent carriers
Testing saliva samples could offer an easy and effective mass testing approach for detecting COVID-19. **http://dgit.in/oct20-37**



COVID-19 affects testosterone
Over half of male patients studied were found to have lower than their normal testosterone levels. **http://dgit.in/oct20-38**

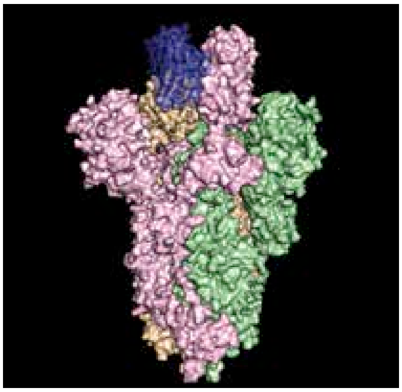
another molecule, that is not a peptide, that can serve the same function.

Researchers from Belgium and the University of Texas have found unlikely help from llama's. These South American mammals produce special types of antibodies in response to infections, which are totally unlike the ones that are say produced by humans. These antibodies are about one fourth the size of the antibodies produced by humans, and bond with the virus directly, preventing them from infecting other cells. The researchers were investigating the use of these antibodies to treat MERS and SARS. The same team investigated if any of the antibodies that were effective against these coronaviruses would be effective against SARS-CoV-2 as well. One of the antibodies tested, SARS VHH-72, showed an ability to bind to the SARS-CoV-2 spike protein as well, but it also unbound quickly. The researchers tweaked the structure to bind to the spike protein of the novel coronavirus more effectively. The researchers now hope to test the treatment on hamsters or primates, and then move on to human clinical trials.

These nanobodies are stable for long periods of time, and can be introduced to the body quickly through the use of inhalers. Unlike vaccines, which take one or two months to build up the immunity of the body, the nanobodies can start protecting the body immediately, and are very helpful for those who are already immunocompromised.

WASTEWATER TESTING

Researchers from Ohio State University are sampling the waste water from the state on its way to treatment plants, to predict escalations in the number of cases. Since the RNA fragments of the virus finds its way to bodily waste before infected people manifest symptoms, studying waste water can allow scientists to get a warning of spikes in cases about a week before they actually take place. A similar approach was used in the University of Arizona, on a smaller scale, to effectively stop an outbreak. After students returned to college, researchers sampled the waste water from the campus every morning. Within the first week of the college reopening, traces of the virus were found in the wastewater. After testing all the students on the campus, two asymptomatic individuals were identified and quarantined, preventing the virus from spreading further. Previously, this approach has been used to monitor polio, antimicrobial resistance, and the overuse of opioids. The establishment in America is working on a nationalised wastewater surveillance system.



The Llama nanobody, SARS VHH-72 (in blue) binds to the spike protein in SARS-CoV-2 (pink, green and orange).

UNDERSTANDING THE PANDEMIC

Researchers from the Purdue University wanted to find out how people responded to various restrictions around the world, to what degree they followed the guidelines, and which policies are the most effective at controlling the spread of the coronavirus. To cover the entire planet, they had to use footage from 30,000 cameras spread out over a 100 countries. Since March, the site, called cam2project.net has pulled the data from the scattered locations, and has reduced the study to a single metric – how much of a distance people keep from each other. The site automatically crawled the internet for publicly available cameras, and created a backup of the data over intervals of 10 minutes. The site can be used to measure how crowded places got after restrictions were eased, and if it was a sudden or gradual increase. There is no use of facial recognition technology in the site, and all the data is anonymised. The feeds from three Indian cities are included in the study – Coimbatore, Delhi and Agra.

Researchers from Caltech have been investigating how the response and attitudes of the public changed, over the course of the outbreak. The findings revealed that during the early days of the pandemic, most people perceived a low risk of possible infection, and underestimated their chances of coming in contact with someone who had the virus, or how seriously sick they would get even if they were infected. With the spread of the virus, the perceived threat increased, and the people followed suit by changing their behaviours including practicing social distancing, washing hands with soap and wearing a mask. This is a kind of optimism bias that has been previously well studied. The study found that information campaigns as well as alerts to smartphones can both be powerful tools in changing the attitudes of people towards the pandemic.

Researchers from the University of Michigan have been studying how the novel coronavirus spreads indoors, studies that could be valuable in helping offices, businesses and schools from limiting the spread of the disease. The researchers used the aerosols generated by healthy people as a proxy for studying the aerosols generated by asymptomatic patients, through activities such as talking or breathing. The research found that the aerosols could stick to surfaces and be inhaled by other individuals in the area. Ample ventilation reduced the amount of viruses in the area, but also led to more viruses sticking to surfaces. In a classroom, after a teacher gave a 50 minute long lecture, only 10 percent of the simulated virus particles were filtered out. The ventilation forms small vortexes that prevent the virus from leaving the area. If the speaker stayed under an air vent, the virus spread increased significantly. The research could help inform how spaces such as conference rooms, theatres and supermarkets are set up and disinfected.